NUMERICAL ANALYSIS
PROBLEM SET 4

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1. Problem 1

1.a. We are deriving the 2-point open Newton-Coats formula

$$\int_a^b f(x)dx \approx I_{2,\text{open}}(f) = \frac{b-a}{3}\Big(2f(a+h) - f(a+2h) + 2f(a+3h)\Big)$$

for evenly spaced points with $h = \frac{b-a}{n+2}$. Now we set $n = 2$ so that $h = \frac{b-a}{4}$, and we need to sort out

$$I_{2,\text{open}}(f) = \int_a^b \sum_{i=0}^{2} L_{2,i}(x) f(x_i) dx. \tag{1}$$

But we need to shift to an easier or simpler variable to work with. So we set $x = a + t \cdot h$, with $dx = h \cdot dt$. Now as x moves from a to b we see that $(b = a + 4h)$, t moves from 0 to 4, and we have

$$\int_a^b f(x)dx = h \int_0^4 f(a+ht)dt,$$

which we interpolate $f(a + ht)$ at $t = 1, 2, 3$. We write out the Lagrangian polynomials $L_{2,0}, L_{2,1}, L_{2,2}$ for the sake of the problem. This follows as

$$\begin{aligned}
L_{2,0}(x) &= \frac{(t-2)(t-3)}{(1-2)(1-3)} = \frac{(t-2)(t-3)}{2} = \frac{1}{2}(t^2 - 5t + 6)\\
L_{2,1}(x) &= \frac{(t-1)(t-3)}{(2-1)(2-3)} = -(t-1)(t-3) = -t^2 + 4t - 3\\
L_{2,2}(x) &= \frac{(t-1)(t-2)}{(3-1)(3-2)} = \frac{(t-1)(t-2)}{2} = \frac{1}{2}(t^2 - 3t + 2).
\end{aligned}$$

Date: April 13, 2026.

And we then need to integrate these Lagrange interpolants, which we do as

$$\int_0^4 \frac{1}{2}(t^2-5t+6)dt = \frac{t^3}{6} - \frac{5t^2}{4} + 3t\Big|_0^4 = \frac{8}{3}$$

$$\int_0^4 -t^2+4t-3dt = -\frac{t^3}{3} + 2t^2 - 3t\Big|_0^4 = -\frac{4}{3}$$

$$\int_0^4 \frac{1}{2}(t^2-3t+2)dt = \frac{t^3}{6} - \frac{3t^2}{2} + 2t\Big|_0^4 = \frac{8}{3}$$

Thus, when we plug this into equation (1), we get the following

$$\begin{aligned}\int_a^b f(x)dx &\approx h\Big(\frac{8}{3}f(a+h) - \frac{4}{3}f(a+2h) + \frac{8}{3}f(a+3h)\Big)\\ &\approx \frac{4h}{3}\Big(2f(a+h) - f(a+2h) + 2f(a+3h)\Big)\\ &\approx \frac{b-a}{3}\Big(2f(a+h) - f(a+2h) + 2f(a+3h)\Big).\end{aligned}$$

which solves our problem

1.b. We must show that our quadrature is exact for $1, x, x^2, x^3$ but not for x^4. Again we rescale the interval in teh fashion

$$x = a + ht, \qquad h = \frac{b-a}{4}, x \in [a,b] \to t \in [0,4]$$

So $Q(g) = \frac{4}{3}\Big(2g(1) - g(2) - 2g(3)\Big)$. But, we need to check if $I(g) = Q(g)$ for $g(t) \in \{1, x, x^2, x^3, x^4\}$. So we write

$$\int_0^4 1dt = 4 \stackrel{?}{=} \frac{4}{3}(2-1+2) = 4 \quad \checkmark$$

$$\int_0^4 tdt = \frac{t^2}{2}\Big|_0^4 = 8 \stackrel{?}{=} \frac{4}{3}(2\cdot 1 - 2 + 2\cdot 3) = 8 \quad \checkmark$$

$$\int_0^4 t^2dt = \frac{t^3}{3}\Big|_0^4 = \frac{64}{3} \stackrel{?}{=} \frac{4}{3}(2\cdot 1 - 4 + 2\cdot 9) = \frac{4\cdot 16}{3} = \frac{64}{3} \quad \checkmark$$

$$\int_0^4 t^3dt = \frac{t^4}{4}\Big|_0^4 = 64 \stackrel{?}{=} \frac{4}{3}(2\cdot 1 - 8 + 2\cdot 27) = \frac{4\cdot 48}{3} = \frac{192}{3} = 64 \quad \checkmark$$

$$\int_0^4 t^4dt = \frac{t^5}{5}\Big|_0^4 = \frac{1024}{5} = 204.8 \stackrel{?}{=} \frac{4}{3}(2 - 32 + 2\cdot 243) = \frac{1824}{3} = 608 \quad \times$$

So we are done.

2. Problem 2

Let

$$I(f) = \int_a^b f(x)dx$$

and let $[a, b]$ be divided into subintervals in the following fashion

$$\big\{[a, x_1 + h], [x_2 - h, x_2 + h], \ldots, [x_n - h, b]\big\},$$

with $x_1, \ldots, x_n$ being the midpoints of the subintervals. We thusly see that $h = \frac{b-a}{2n}$, that $x_j = a + (2j - 1)h$, and that each of our subintervals has width $2h$. Now consider the j^{th} subinterval $[x_j - h, x_j + h]$. We can write $f(x)$ over the interval using Taylor's theorem as

$$f(x) = f(x_j) + f'(x_j)(x - x_j) + \frac{f''(\eta_x)}{2}(x - x_j)^2,$$

for some η_x between x and x_j. So we integrate

$$\begin{aligned}
\int_{x_j-h}^{x_j+h} f(x)dx &= \int_{x_j-h}^{x_j+h} \left(f(x_j) + f'(x_j)(x - x_j) + \frac{f''(\eta_x)}{2}(x - x_j)^2 \right) dx \\
&= \int_{x_j-h}^{x_j+h} f(x_j)dx + \int_{x_j-h}^{x_j+h} f'(x_j)(x - x_j)dx + \int_{x_j-h}^{x_j+h} \frac{f''(\eta_x)}{2}(x - x_j)^2 dx \\
&= 2hf(x_j) + 0 + \frac{f''(\xi_j)}{2} \int_{x_j-h}^{x_j+h} (x - x_j)^2 dx,
\end{aligned}$$

for some ξ_j clearly dependent upon our choice of η_x. But notice that our last integral can be manipulated by a transformation of variables in the following fashion

$$\int_{x_j-h}^{x_j+h} (x - x_j)^2 dx = \int_{-h}^{h} t^2 dt = \frac{2h^3}{3}.$$

Substituting that back in to the over all equation we see that

$$\int_{x_j-h}^{x_j+h} f(x)dx + \frac{f''(\xi_j)}{\cancel{2}} \frac{\cancel{2}h^3}{3}$$